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LAMP: Long-Tailed Multi-modal Prompt Tuning for Vision-Language Models

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Abstract

Prompt Learning (PL) has sparked growing interest as an efficient method for adapting large Vision-Language Models (VLMs) to downstream tasks. While most existing PL methods are designed for (nearly) balanced data, real-world datasets always exhibit a long-tailed distribution, which necessitates the design of PL methods specifically for imbalanced scenarios. This paper introduces our LAMP framework to enable VLMs better to learn from the long-tailed base classes and achieve non-biased predictions for both base and new classes. LAMP is integrated into specific intermediate layers of the frozen VLM, where for each layer we introduce three co-designed mechanisms: Multi-modal Prompt Pool (MPP), Modality-Shared Prompts (MSP) and Load Balancing Optimization (LBO). MPP is motivated by the goal of boosting feature clustering to foster the mutual learning of head and tail classes, in which we introduce separated multi-modal prompts that are dynamically combined into the model via similarity metrics. In addition to the prompts retrieved by MPP, we introduce globally shared MSPs to better adapt to cross-modal semantics and enhance the training robustness. Furthermore, to promote tight and discernible feature clustering via prompts, we treat each prompt as an expert and adopt a load-balancing technique from mixture-of-experts in LLMs, named LBO. LBO dynamically adjusts attention weights with an externally optimized bias, thereby making the activation of each prompt more evenly distributed and preventing the overfitting of head classes. Extensive experiments under various long-tailed settings demonstrate that our LAMP consistently outperforms other state-of-the-art methods.

Keywords Vision-language models · Prompt learning · Long-tailed learning

1 Introduction

Recent years have witnessed a striking advancement in Vision-Language Models (VLMs) (Radford et al., 2021; Jia et al., 2021; Li et al., 2022), which has demonstrated unprecedented modeling and generation abilities by training over web-scale image-text corpora. As one of the pioneering VLMs, CLIP (Radford et al., 2021) is pre-trained over 400-million unlabeled image-text pairs with contrastive learning to establish a modality-joint feature space that captures cross-modal correspondences; Supported by the comprehensiveness and generality, it can be easily adapted to various downstream tasks, i.e., cross-modality generation (Ge et al., 2023; Chen et al., 2024; Zhang et al., 2021), image captioning (Barraco et al., 2022; Kornblith et al., 2023), retrieval (Baldrati et al., 2022; Zhang et al., 2024a), vision question answering (Keskar et al., 2025) and etc.

To efficiently adapt VLMs to specific datasets, an emerging trend is to conduct Prompt Learning (PL) (Lu et al., 2022; Zhou et al., 2022b, a; Zhang et al., 2024b; Shi et al.,

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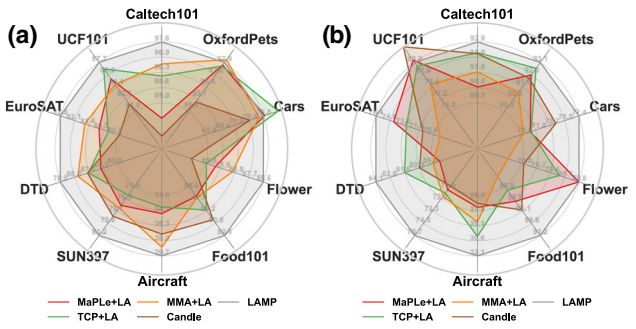


Fig. 1 LAMP achieves significant performance gains in harmonic mean of base/new accuracy (**a**) and base/new macro-F1 (**b**) on the long-tailed base-to-new benchmark

2024b; Xu et al., 2025), which introduces several trainable parameters plugged into frozen VLMs to register task-specific knowledge. Early prompt learning methods, like CoOp (Zhou et al., 2022b), CoCoOp (Zhou et al., 2022a), learn isolated/conditioned textual prompt templates to guide task-aware modality alignment; MaPLe (Khattak et al., 2023a) extends this idea by incorporating layer-wise prompting, where textual prompts are employed for each layer and utilized to generate visual prompts. To avoid catastrophic forgetting caused by excessive prompt tuning, some methods (Zhu et al., 2023; Khattak et al., 2023b) specifically design self-supervision or gradient refinement techniques, where models are encouraged to narrow the discrepancy between the unbiased zero-shot CLIP. Recent methods, like MMA (Yang et al., 2024) and MMRL (Guo & Gu, 2025a, b), further enhance the adaptation performance by explicitly modeling cross-modal interactions or integrating multiple tuning strategies.

Notably, most existing PL methods (Zhou et al., 2022b, a; Xu et al., 2023; Yang et al., 2024; Guo & Gu, 2025a, b) for VLMs are designed for (nearly) balanced multi-modal datasets, which deviates from real-world scenarios where data is universally long-tail distributed (Du et al., 2024b, a). This highlights that PL on VLMs should also be specifically tailored to these circumstances. Unfortunately, few methods have discussed this: some existing studies (Jia et al., 2022; Dong et al., 2022a; Xu et al., 2024) propose visual prompt finetuning under long-tailed scenarios, but only for vision models; The method most closely related to our work is Candle (Shi et al., 2024a), which employs visual/textual prototypes as well as cross-modality attention to encourage balanced recognition of multi-modal datasets. However, it only discusses the output-wise balancing techniques to facilitate the construction of prototypes for classification; *further insights on how refined prompts can help with the long-tailed problem, as well as utilizing modality interactions to achieve balanced learning, remain unexplored*.

Contributions

To alleviate the above issues, this paper introduces a novel Long-tailed Multi-modal Prompt learning (dubbed LAMP) approach. Our motivation stems from observations in traditional long-tailed (LT) learning (Li et al., 2023, 2022; Du et al., 2024a, b): despite differences in form, most methods aim to promote the formation of compact, cross-category feature clusters in the embedding space, thereby overcoming performance bias caused by class size. Following this philosophy of feature clustering, we propose to use different prompts to cluster features in long-tailed multi-modal datasets, boosting the mutual learning of head and tail classes during prompt fine-tuning. This leads to our design of the Multi-modal Prompt Pool (MPP), which maintains a set of image-text prompt pairs. For each instance, we use its [CLS] token to retrieve from the MPP and select prompts based on feature similarity. Furthermore, aware of the peculiarity of multi-modal learning, we concatenate the retrieved prompts with Modality-Shared Prompts (MSPs) to better capture cross-modal semantics. However, relying solely on spontaneous feature clustering can still lead to overfitting of head-class features. Inspired by load balancing in Mixture-of-Experts (MoE) layers of Large Language Models, we further propose Load Balancing Optimization (LBO) to equalize the activation levels across different prompts. Applied during prompt querying, LBO first accumulates the retrieval rankings of each prompt across the dataset. Then, we introduce an externally optimized bias term to readjust the activation weights of prompts over the entire dataset. By optimizing this bias, we encourage the prompts to capture class-discriminative, compact, and evenly sized feature manifolds at the dataset level, thereby facilitating long-tailed learning. Extensive experiments on 11 LT-B2N datasets (15 datasets in total, including ImageNet OOD variants) under imbalanced settings demonstrate the consistent superiority of LAMP over existing methods; as shown in Fig. 1, LAMP achieves significant improvements in both accuracy and macro-F1. We summarize our contribution as follows:

- To the best of our knowledge, we present the first prompt-based PL framework for long-tailed multi-modal VLMs, complementing adapter-centric methods such as Candle (Shi et al., 2024a) and offering greater applicability and practical significance compared with existing balanced PL methods.
- We propose the LAMP framework, which incorporates three co-designed mechanisms to enhance feature clustering and mutual learning between head and tail classes: the Multi-modal Prompt Pool (MPP), Modality-Shared Prompts (MSPs), and Load Balancing Optimization (LBO).

- Extensive experiments on 11 LT-B2N datasets and additional OOD benchmarks under various imbalance settings demonstrate that LAMP consistently outperforms competing methods in both accuracy and class re-balancing capability. Further ablation studies confirm that LAMP is computationally efficient and the three mechanisms operate synergistically.

2 Related Works

2.1 Prompt Learning of VLMs

The introduction of prompt learning on VLMs is built on its wide application across individual modalities (Jia et al., 2022; Li et al., 2024; Jiang et al., 2022; Dong et al., 2022b). Early methods like CoOP (Zhou et al., 2022b) and CoCoOp (Zhou et al., 2022a) only add prompts in the text encoder. MaPLe (Khattak et al., 2023a) proposes to combine prompts in each layer of text encoders, and generate corresponding layer-wise image prompts from them; ProDA (Lu et al., 2022) and ProGrad (Zhu et al., 2023) discuss the supervision scheme of obtaining the optimal text prompts. ProVP (Xu et al., 2025) inserts layer-wise prompts into the forward flow in an incremental manner. DPL (Xu et al., 2023) decouples the attention within/between tokens and prompts, respectively, and finds that token-prompt cross-attention contributes most to the performance. DePT (Zhang et al., 2024b) raises a channel-adaptive classifier for prompt learning to mitigate the channel bias of domain generalization. (Shi et al., 2024b) explores three correlation types between tokens and prompts when learning with missing modalities. The recent work MMRL (Guo & Gu, 2025a, b) proposes to insert shared prompts into high layers for fine-tuning, and further integrates multiple self-supervised methods as described above to achieve optimal performance. There are also theoretical studies, for example, (Wang et al., 2023; Hu et al., 2024) justify the theoretical limitations of prompt learning compared from the perspective of convex cones.

2.2 Long-Tailed Learning

Real-world datasets follow a prevalent Long-Tailed (LT) distribution, where easy-to-acquire samples are always multiple times than rare ones. Typical LT solutions lie in reweighting (Menon et al., 2020; Cui et al., 2019; Cao et al., 2019; Ren et al., 2020; Tan et al., 2020), resampling (Kubat et al., 1997; Wallace et al., 2011; Chawla et al., 2002), contrastive learning (Zhu et al., 2022; Du et al., 2024b) and etc. For example, ProCo (Du et al., 2024b) borrows the mixture of von Mises-Fisher (vMF) distribution to model the long-tailed contrastive learning. Recently, some LLM-driven solutions have emerged; for example, LTGC (Zhao et al., 2024) leverages LLM-generated content to enrich tail representations; LPT (Dong et al., 2022a) introduces a group tuning technique over the vision models to alleviate the long-tailed effects. Candle (Shi et al., 2024a) explores the long-tailed multi-modal dataset, and proposes an adapter tuning technique accompanied by logit-adjustment and class-prototypes. *In contrast to them, our study brings novel insights into further utilizing multi-modal prompts over VLMs to achieve balanced learning of multi-modal datasets*.

3 Methodology

3.1 Preliminaries

We first introduce the general settings of prompt learning on CLIP. Define the image and text branches of CLIP as $\mathcal{V}$ and $\mathcal{T}$ where both of them yield N layers, i.e., $\mathcal{V} = \{V_i\}_{i=1}^{N}$ and $\mathcal{T} = \{T_i\}_{i=1}^{N}$; for a multi-modal sample pair $\{\boldsymbol{x}, y\}$ from the minibatch $\{\mathcal{X}, \mathcal{Y}\}$, we denote B as the batchsize, $\boldsymbol{x} \in \mathbb{R}^{H\times W\times 3}$ denotes the image, t denotes the input text, y denotes the label. We additionally use $\mathcal{C}$ to denote all classes. The image branch first divides x into N_v size-fixed patches, yielding patch embedding $E \in \mathbb{R}^{N_v\times D_v}$ (D_v is the visual hidden dimension). Later, E is jointly fed into cascaded encoder layers along with a fixed class embedding cls. The text branch first pads the input text t with a predefined template (i.e., "a photo of a [class]") and tokenizes it, yielding $T \in \mathbb{R}^{N_t\times D_t}$ (D_t is the textual dimension). Then, with the Begin-Of-Text (BOT) bot and the End-Of-Text (EOT) eot as indicators, T is processed with cascaded $\{\mathcal{T}_i\}$. The process of each image and text encoder layer can be formalized as a recurrent form with the layer number omitted:

$$\begin{aligned} [E, cls] &\leftarrow V([E, cls]); \\ [bot, T, eot] &\leftarrow T([bot, T, eot]) \end{aligned} \tag{1}$$

The visual and text output of the last layer, represented by cls_N, eot_N, are projected into a shared space with $P_v \in \mathbb{R}^{D_v\times D_s}$ and $P_t \in \mathbb{R}^{D_t\times D_s}$ (D_s is the dimension of shared output space):

$$F = P_v(cls_N); \quad W = P_t(eot_N) \tag{2}$$

For the final classification stage, CLIP uses the final visual output F and calculates its similarity with text outputs $\{W_c\}_{c=1}^{C}$ for each class to generate prediction $\boldsymbol{p}(y_c)$, i.e:

$$\boldsymbol{p}(y_c) = \sigma(\mathtt{sim}(F, W_c)) = \frac{\exp(\mathtt{sim}(F, W_c))}{\sum_{i=1}^{C}\exp(\mathtt{sim}(F, W_i))} \tag{3}$$

where σ means the softmax. $\mathtt{sim}(\cdot,\cdot)$ calculates the cosine similarity. For the training stage, $\boldsymbol{p}(y_c)$ is pushed to learn the label y with the cross-entropy loss $\mathcal{L}_{CE}$.

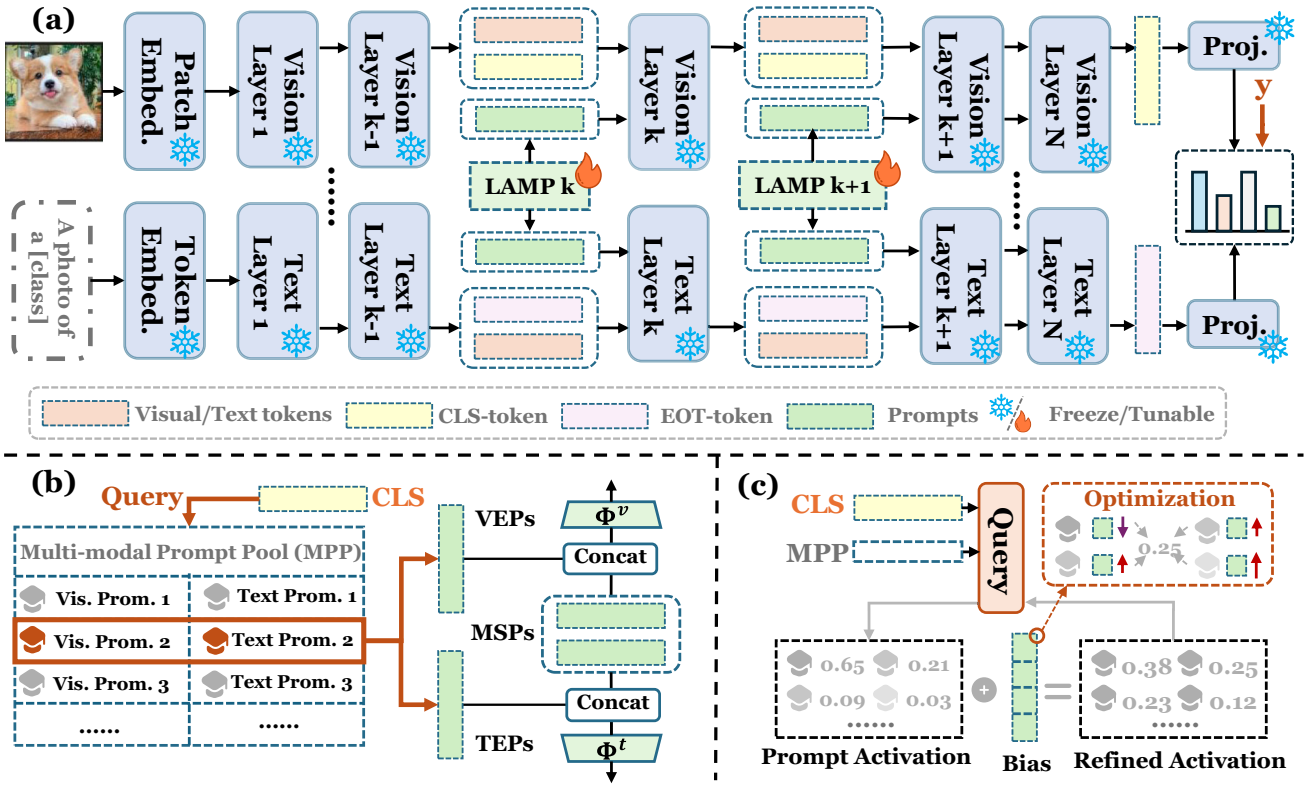


Fig. 2 **a** The overview of LAMP, where we leverage prompts generated from LAMP modules in specific layers to dynamically balance the long-tailed effects. Each module yields three key designs: Multi-modal Prompt Pool (MPP), Modality-Shared Prompts (MSP) and Load Balancing Optimization (LBO). **b** We query MPP with cls-token and concat the retrieved VEPs/TEPS with our MSPs. **c** LBO applied in the query operation. We optimize the extra bias to enforce even activations of different prompts

3.2 LAMP: Long-Tailed Multi-modal Prompt learning

This section introduces our framework, consisting of cascaded LAMP modules plugged into the visual and text encoders. Each LAMP module comprises three key designs: *Multi-modal Prompt Pool (MPP)*, *Modality-Shared Prompts (MSP)* and *Load Balancing Optimization (LBO)*.

3.2.1 Multi-modal Prompt Pool

A key principle underlying traditional long-tail (LT) methods is to promote feature clustering across samples and classes, thereby overcoming the performance bias caused by class size and enhancing the learning of both head and tail classes. Similarly, in our prompt learning setting, we aim to leverage prompts to dynamically form consistent and distinct clusters across the entire long-tailed dataset. A straightforward approach to leveraging prompts for feature clustering is to partition the dataset into multiple clusters based on feature similarity, and then assign distinct prompts to each cluster. Based on this motivation, we maintain a *Multi-modal Prompt Pool (MPP)* $\mathcal{S}$ at each selected layer. Concretely, $\mathcal{S}$ contains M prompt pairs $\{(\boldsymbol{v}_1, \boldsymbol{t}_1), (\boldsymbol{v}_2, \boldsymbol{t}_2), ..., (\boldsymbol{v}_M, \boldsymbol{t}_M)\}$ where $\boldsymbol{v}_m \in \mathbb{R}^D$ is the visual prompt and $\boldsymbol{t}_m \in \mathbb{R}^D$ is the text prompt. The pool is layer-specific; the retrieved prompts are not pre-computed features for offline matching. Instead, after retrieval they are inserted as prompt tokens into the current visual and textual transformer layers and participate in the subsequent attention computation. In this way, both the query and the injected prompt tokens reside in the same layer, which avoids directly matching representations from different semantic levels. For each input $\boldsymbol{x}$, we obtain its cls-token $cls_i(\boldsymbol{x})$ in the current layer i, and then use this token to retrieve K prompts from the prompt pool $\mathcal{S}$:

$$\begin{aligned} &\mathcal{I}_i(\boldsymbol{x}) = \mathcal{T}(\{\texttt{sim}(cls_i(\boldsymbol{x}), \boldsymbol{v}_{m'})\}_{m'=1}^{M}; K), \\ &\texttt{Ret} = \{(\boldsymbol{v}_m, \boldsymbol{t}_m) \mid m \in \mathcal{I}_i(\boldsymbol{x})\}. \end{aligned} \tag{4}$$

In the following retrieval equations, $\mathcal{T}(x; K)$ denotes the top-K selection operator, returning the indices of the K largest entries in x. LAMP performs visual-query retrieval of paired visual–text prompts rather than retrieval with a text query. The visual cls-token is the sample-specific query available without using the ground-truth class label; it determines the top-K indices, and both $\boldsymbol{v}_m$ and $\boldsymbol{t}_m$ at those indices are selected for injection into the visual and textual branches,

respectively. This design preserves instance-specific retrieval from the visual branch while maintaining the paired multi-modal structure of MPP (Fig. 2). Table 8 studies the effect of injecting different retrieved prompt modalities under the same visual-query retrieval mechanism. We then concatenate these retrieved prompts separately by modality, generating the Vision Extra Prompts (VEPs) $\boldsymbol{v}_{\text{ex}}$ and Text Extra Prompts (TEPs) $\boldsymbol{t}_{\text{ex}}$:

$$\begin{aligned} \boldsymbol{v}_{\text{ex}} &= \texttt{Concat}(\{\boldsymbol{v}_m \mid m \in \mathcal{I}_i(\boldsymbol{x})\}); \\ \boldsymbol{t}_{\text{ex}} &= \texttt{Concat}(\{\boldsymbol{t}_m \mid m \in \mathcal{I}_i(\boldsymbol{x})\}) \end{aligned} \tag{5}$$

In the following, we introduce another prompt that works synergistically with VEPs and TEPs to better capture cross-modal knowledge.

3.2.2 Modality-Shared Prompts

Inspired by previous studies (Khattak et al., 2023a; Guo & Gu, 2025a) showing that cross-modal prompts are more effective at capturing multi-modal semantics, we introduce additional *Modality-Shared Prompts (MSP)* to further enhance the perception of cross-modal semantics. Let $\boldsymbol{s} \in \mathbb{R}^{S\times D}$ be the MSP, where S denotes its length. Then, $\boldsymbol{s}$ is concatenated separately with $\boldsymbol{v}_{\text{ex}}$ and $\boldsymbol{t}_{\text{ex}}$, and processed by separate projectors ϕ^v and ϕ^t to yield P^v and P^t:

$$\begin{aligned} P^v &= \phi^v(\texttt{Concat}(\boldsymbol{v}_{\text{ex}}, \boldsymbol{s})); \\ P^t &= \phi^t(\texttt{Concat}(\boldsymbol{t}_{\text{ex}}, \boldsymbol{s})); \end{aligned} \tag{6}$$

P^v and P^t are then inserted into the forward process of the current layer. We can reformulate Eq. 1 as the following:

$$\begin{aligned} [E, P^v, cls] &\leftarrow V([E, P^v, cls]); \\ [bot, P^t, T, eot] &\leftarrow T([bot, P^t, T, eot]) \end{aligned} \tag{7}$$

In the preceding sections, we introduced how to achieve preliminary balanced learning through a dynamic multi-modal prompt pool (MPP) together with MSPs. In practice, each prompt in MPP operates similarly to an expert in a mixture-of-experts architecture, selectively activated at specific steps via similarity-based queries. However, as noted by Jiang et al. (2025), Xiao et al. (2025), this paradigm can lead to excessive sensitivity to initialization and overloading of a small subset of experts. In the next section, we introduce a loss-level innovation termed LBO to proactively enable efficient collaboration among experts and capture more cross-category patterns.

3.2.3 Load Balancing Optimization

In the previous sections, we promote feature re-clustering and interaction by dynamically assigning different prompts to features. However, allowing the model to spontaneously learn feature clustering from data still tends to produce tighter manifolds for head-class features. To further enhance cluster distinction, inspired by the load balancing mechanism (Wang et al., 2024; Qiu et al., 2025) in Mixture-of-Experts (MoE) of Large Language Models (LLMs), we propose *Load Balancing Optimization (LBO)*. Specifically, we treat each prompt in the pool as an expert, and the accumulated selection probability of each expert is regarded as its activation value. Our goal is to encourage a consistent and uniform distribution of expert activations across all samples, thereby improving the balance and separation of the features captured by each expert. For each selected layer, LBO maintains a prompt-level bias vector $\boldsymbol{b} \in \mathbb{R}^M$, where the m-th entry is a scalar bias for the m-th prompt pair. This vector has the same length as the similarity score vector computed over the M prompt experts. In our implementation, we modify the querying operation in Eq. 4 as:

$$\begin{aligned} &\boldsymbol{r}_{i,m}(\boldsymbol{x}) = \texttt{sim}(cls_i(\boldsymbol{x}), \boldsymbol{v}_m) + b_m, \quad m = 1, \ldots, M, \\ &\mathcal{I}_i(\boldsymbol{x}) = \mathcal{T}(\{\boldsymbol{r}_{i,m}(\boldsymbol{x})\}_{m=1}^{M}; K), \\ &\texttt{Ret} = \{(\boldsymbol{v}_m, \boldsymbol{t}_m) \mid m \in \mathcal{I}_i(\boldsymbol{x})\}. \end{aligned} \tag{8}$$

We rely on $\boldsymbol{b}$ to explicitly adjust the load of each expert (i.e., prompt). Notably, $\boldsymbol{b}$ does not participate in the autograd optimization, but is updated by an additional process at each step t:

$$b_m \leftarrow b_m - \gamma \cdot \texttt{sign}(\overline{a}_{m,t}(K) - K/M). \tag{9}$$

Recall that K denotes the number of prompts retrieved from the prompt pool. `sign` is the sign function, and γ is the learning rate. Following (Wang et al., 2024), we adopt a cosine scheduler to adjust γ, thereby reducing its variation as the model converges. In the ablation studies, we examine the effect of the initial value of γ. $\overline{a}_{m,t}(K)$ represents the activation frequency of expert m up to step t, updated iteratively based on how often expert m appears in the top-K retrieval set:

$$\begin{aligned} \overline{a}_{m,t+1}(K) =& \frac{G-1}{G}\overline{a}_{m,t}(K) + \\ & \frac{1}{G|\mathcal{X}|}\sum_{\boldsymbol{x}\in\mathcal{X}} \gg 1(m \in \mathcal{I}_i(\boldsymbol{x})), \\ \mathcal{I}_i(\boldsymbol{x}) =& \mathcal{T}(\{\texttt{sim}(cls_i(\boldsymbol{x}), \boldsymbol{v}_{m'}) + b_{m'}\}_{m'=1}^{M}; K). \end{aligned} \tag{10}$$

Here G denotes the number of updates, and $\gg 1(\cdot)$ is the indicator function (1 if the condition holds, 0 otherwise). This update mechanism tracks the load distribution across the dataset. The counter is reset to zero at the start of each epoch. The target load for each expert is K/M, since K experts are retrieved from M candidates per sample. Consequently, Eq. 9

Algorithm 1 LBO update for one selected layer

Input: mini-batch $\mathcal{X}$, prompt pairs $\{(\boldsymbol{v}_m, \boldsymbol{t}_m)\}_{m=1}^{M}$, top-$K$, bias $\boldsymbol{b} \in \mathbb{R}^M$
for each $\boldsymbol{x} \in \mathcal{X}$ **do**
  Compute layer-wise query token $cls_i(\boldsymbol{x})$
  Compute scores $r_{i,m} = \texttt{sim}(cls_i(\boldsymbol{x}), \boldsymbol{v}_m) + b_m$ for all m
  Retrieve $\mathcal{I}_i(\boldsymbol{x}) = \mathcal{T}(\{r_{i,m}\}_{m=1}^{M}; K)$
  Inject $\{(\boldsymbol{v}_m, \boldsymbol{t}_m) \mid m \in \mathcal{I}_i(\boldsymbol{x})\}$ with MSP into the current layer
end for
Update $\overline{a}_{m,t}(K)$ by the retrieval frequencies in the mini-batch
Update $b_m \leftarrow b_m - \gamma \cdot \texttt{sign}(\overline{a}_{m,t}(K) - K/M)$

reduces the bias of over-activated experts and increases the relative chance of under-activated ones, thereby encouraging more balanced activations across different prompts.

3.2.4 Learning Objectives

In this part, we summarize the training process of LAMP. For the prompts retrieved from MPP, we maximize their similarity with the query to encourage the formation of compact and well-separated feature clusters in the dataset. This loss function is combined with the Cross-Entropy Loss (applied on the final outputs $\boldsymbol{p}$) as the following:

$$\mathcal{L} = \mathcal{L}_{\texttt{CE}} + \frac{\beta}{|\texttt{Sec}|} \sum_{i \in \texttt{Sec}} \left(1 - \frac{1}{K} \sum_{k \in \mathcal{I}_i(\boldsymbol{x})} \texttt{sim}(cls_i(\boldsymbol{x}), \boldsymbol{v}_k) \right) \tag{11}$$

where β is a hyper-parameter for controlling the loss contributions; $\texttt{Sec}$ is the selected layer for adding our LAMP module. The similarity loss is averaged over the selected layers and the retrieved prompts in each layer. Additionally, we further employ the LBO introduced above to balance the load distribution of different prompts.For the long-tailed base-to-new benchmark, LAMP first learns prompts from labeled base-class training data. We report this purely inductive variant, denoted as LAMP without calibration, as the direct comparison with inductive baselines. Before evaluating new-class test images, LAMP can optionally calibrate the LBO bias with unlabeled training images from the new classes. No new-class labels or test images are used in this calibration; this optional variant is denoted as LAMP with calibration.

4 Experiments

4.1 Settings

4.1.1 Benchmarks

Following previous approaches, we employ four well-known benchmarks but manipulate them to follow the long-tailed settings:

❶ **Long-Tailed Base-to-New (B2N) Benchmark (LT-B2NB)**, which separates the whole dataset into disjoint base to new classes. Models are trained with labeled base-class training sets, and evaluated on test sets of both base and new classes; for this dataset, we employ 2 object recognition datasets ImageNet (Deng et al., 2009) and Caltech101 (Fei-Fei et al., 2004); 1 scene understanding dataset named SUN397 (Xiao et al., 2010); 1 texture dataset named DTD (Cimpoi et al., 2014); 1 satellite dataset EuroSAT (Helber et al., 2019), 1 action dataset named UCF101 (Soomro et al., 2012) and 5 fine-grained recognition datasets: OxfordPets (Parkhi et al., 2012), StanfordCars (Krause et al., 2013), Flowers102 (Nilsback & Zisserman, 2008), Food101 (Bossard et al., 2014) and FGVC-Aircraft (Maji et al., 2013). The training data is manipulated into long-tailed distribution by down-sampling the base classes.
❷ **Few-Shot Generalization Benchmark (FSGB)**, which assesses the generalization capability of the model under few-shot (default: 16-shot) settings. This benchmark employs the same dataset as B2N but without the long-tailed manipulation;
❸ **Long-Tailed Cross-Dataset Generalization Benchmark (LT-CDGB)**, which assesses the generalization abilities of models across various datasets. The models are first trained on long-tailed source dataset (ImageNet), then evaluated on the balanced target datasets (10 fine-grained datasets introduced in the above);
❹ **Long-tailed Domain Generalization Benchmark (LT-DGB)**, which assesses the generalization capability of models across different domains. The models are first trained on long-tailed ImageNet, then evaluated on four out-of-distribution variants of ImageNet, i.e., ImageNet-A (Hendrycks et al., 2021b), ImageNet-S (Wang et al., 2019), ImageNet-R (Hendrycks et al., 2021a) and ImageNetV2 (Recht et al., 2019).

As for the concrete steps for applying long-tailed distributions to the dataset, we conduct down-sampling to manipulate the size of all classes into an exponential decayed long-tail curve (Shi et al., 2024a) satisfying a certain imbalance ratio $\texttt{imb}$ (if the sample number of class i is $|\mathcal{C}_i|$, the imbalance ratio is calculated with $\texttt{imb} = \texttt{max}(|\mathcal{C}_i|)/\texttt{min}(|\mathcal{C}_i|)$). In down-sampling, we fix the $max(|\mathcal{C}_i|)$ as the size of the maximum class in the original data and apply different imbalance ratios. By default, we treat the top 20% classes and others as tail classes (Table 1).

Baselines

We compare LAMP with typical prompt learning methods like CoOp (Zhou et al., 2022b), CoCoOp (Zhou et al., 2022a), LFA and MaPLe (Khattak et al., 2023a); We also consider several recent state-of-the-art methods, includ-

Table 1 Statistics for datasets used in our experiments.

Dataset	Classes	Train	Test	oLT
ImageNet	1000	1.28M	50000	×
Caltech101	100	4128	2456	✓
OxfordPets	37	2944	3669	×
StanfordCars	196	6509	8041	×
Flowers102	102	4093	2463	✓
Food101	101	50500	30300	×
FGVCAircraft	100	3334	3333	×
SUN397	397	15880	19850	×
DTD	47	2820	1692	×
EuroSAT	10	13500	8100	✓
UCF101	101	7639	3783	✓
ImageNet-A	200	\	7500	✓
ImageNetV2	1000	\	10000	×
ImageNet-S	1000	\	50889	×
ImageNet-R	200	\	30000	✓

Here oLT means whether the dataset is originally long-tailed

ing Candle (Shi et al., 2024a), TCP (Yao et al., 2024) and MMA (Yang et al., 2024). Notably, since these methods are not crafted for long-tailed circumstances except for Candle, we follow Shi et al. (2024a) to adopt a Logit-Adjusted Loss (Menon et al., 2020) to ensure fairness. All methods are evaluated under the same class split, imbalance ratio, shot number, and random seeds in each benchmark whenever the released implementation supports the same protocol. LFA is only compared in the first two benchmarks due to its limited compatibility with other settings. More details are in Appx. B.

Implementation Details

We use a pretrained CLIP with ViT-B/16 for all datasets. For the main long-tailed B2N results in Tables 2 and 3, we use AdamW with a learning rate of 3×10^{-4} and a weight decay of 5×10^{-4}. The training and inference batch sizes are 32 and 100, respectively. Models are trained for 50 epochs with 16 shots per class under imbalance ratios $\texttt{imb} \in \{10, 20, 50\}$. The default prompt dimension is 64, and LAMP modules are inserted into layers 6–12. The default prompt configuration is $M = 20$, $K = 6$, $S = 4$, and $\beta = 2$. Results are averaged over five independent seeds, and the standard deviations for LAMP are reported separately in Appendix Table 18.

4.2 Experiment Results

4.2.1 Results on the LT-B2NB

We report the comparison results in Tables 2 and 3. For fairness, the tables now separate the purely inductive variant LAMP without calibration from the optional calibrated variant LAMP with calibration. LAMP without calibration already achieves competitive or leading results under the same inductive protocol as the baselines, with average HM-A/HM-M of 80.00/74.49, 78.66/73.27, and 78.12/72.01 under imbalance ratios 10, 20, and 50, respectively. LAMP with calibration further improves these averages to 80.93/75.66, 79.73/74.57, and 78.79/72.76 by calibrating only the LBO bias with unlabeled new-class training images. The corresponding standard deviations of LAMP under the same B2N protocol are provided in Appendix Table 18. Also, we find that the performance gap between text-only methods CoOp / CoCoOp and others is narrowed with the increment of the imbalance ratio, and the performance degradation of MaPLe, and MMA is especially considerable. This highlights that text information is crucial for keeping the balanced performance on long-tailed datasets. We also observe that LAMP yields a greater performance gain in Macro-F1 than in accuracy, particularly as the imbalance setting worsens. This indicates that our model is more adept at balancing the acquired knowledge under extreme conditions than other methods.

4.2.2 Results on the FSGB

To further show the robustness of LAMP and also align with most previous methods, we report the results of few-shot (16-shot) benchmarks in Table 4. We find that under the few-shot setting, LAMP still demonstrates superior performance: although it performs suboptimally in several datasets like Caltech101, DTD and EuroSAT, its averaged accuracy on base and new classes surpass the previous state-of-the-art by approximately 1.10% and 0.56% respectively. This further validates the strong robustness of LAMP in extreme scenarios.

4.2.3 Results on the LT-CDGB

The experimental results in Table 5 reveal several compelling findings regarding the efficacy of our proposed LAMP framework. LAMP achieves consistent gains across source and target datasets, which indicates that the proposed prompt allocation and balancing strategy does not overfit to the source distribution. This significant margin underscores the effectiveness of our architectural design and its optimization strategy.

Moreover, the robustness of LAMP is particularly evident when applied to challenging and less common datasets. On specialized benchmarks such as DTD and EuroSAT, the model yields considerable performance gains. Notably, these datasets represent visual domains that are distinct from the conventional object-centric imagery prevalent in standard training corpora. The superior performance of LAMP in

Table 2 Results on long-tailed B2N benchmarks with an imbalance ratio of 10 / 20 averaged over 5 runs

Method | imb = 10	Caltech101		OxfordPets		Stanfordcars		Flower102		Food101		FGVC-Aircraft	
	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M
CoOp+LA	91.78	90.96	94.10	90.44	69.23	66.44	71.86	64.42	89.25	88.83	31.12	23.41
CoCoOp+LA	95.09	91.23	96.69	91.91	71.71	67.19	77.61	73.68	91.20	89.02	33.71	26.15
LFA	95.69	90.72	94.09	89.60	72.72	68.72	84.38	78.16	90.44	88.64	34.27	27.76
MaPLe+LA	95.00	90.60	97.64	92.46	76.65	72.40	86.24	80.80	88.36	87.74	35.98	28.22
TCP+LA	96.43	92.31	97.60	92.31	**78.80**	75.01	85.78	78.66	89.79	88.60	35.41	27.90
MMA+LA	96.84	92.36	**97.88**	92.80	75.75	72.48	87.09	80.04	88.49	86.63	38.93	30.56
Candle	94.40	91.37	95.99	91.59	74.30	71.69	85.03	78.05	90.80	85.29	37.78	29.40
LAMP without calibration	96.86	92.50	97.30	92.69	76.41	77.09	87.94	80.05	91.67	90.22	38.45	29.75
LAMP with calibration	**97.58**	**92.91**	97.76	**93.09**	76.45	**77.37**	**88.65**	**80.81**	**92.45**	**91.15**	**40.69**	**32.13**

Method | imb = 10	SUN397		DTD		EuroSAT		UCF101		ImageNet		✦ Average ✦	
	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M
CoOp+LA	72.15	68.38	54.88	40.97	54.42	49.84	64.74	63.32	68.05	62.45	69.23	64.50
CoCoOp+LA	77.99	70.98	65.11	56.35	60.28	54.48	76.78	72.91	70.19	64.38	74.21	68.93
LFA	78.39	72.44	67.43	60.77	69.56	57.79	82.71	78.42	70.44	64.80	76.37	70.71
MaPLe+LA	78.16	72.21	67.20	59.60	81.29	74.80	85.90	82.27	71.78	65.30	78.56	73.31
TCP+LA	77.15	71.92	68.07	61.03	80.94	72.24	87.00	83.79	70.24	64.19	78.84	73.45
MMA+LA	78.43	72.46	68.80	62.07	81.84	73.68	85.30	81.61	69.15	63.77	78.95	73.50
Candle	79.26	72.70	68.13	60.07	80.51	70.72	83.17	79.52	68.89	62.10	78.02	72.05
LAMP without calibration	80.47	74.40	69.88	63.95	82.07	70.73	86.79	82.00	72.18	65.99	80.00	74.49
LAMP with calibration	**81.18**	**75.66**	**70.02**	**64.10**	**84.78**	**76.36**	**87.70**	**82.02**	**72.96**	**66.67**	**80.93**	**75.66**

Method | imb = 20	Caltech101		OxfordPets		Stanfordcars		Flower102		Food101		FGVC-Aircraft	
	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M
CoOp+LA	92.65	90.92	94.15	90.68	67.39	62.31	73.72	67.61	86.38	86.08	29.33	23.16
CoCoOp+LA	95.25	92.49	96.64	92.08	71.38	70.32	80.31	72.42	91.20	88.42	32.78	26.44
LFA	95.56	92.12	90.90	86.45	70.35	68.36	84.03	78.51	89.72	85.92	33.02	27.05
MaPLe+LA	96.09	92.51	96.29	92.19	74.88	71.20	87.84	81.10	90.54	87.98	34.88	28.07
TCP+LA	97.78	93.76	95.75	92.01	73.34	70.45	87.27	80.98	89.90	86.40	36.26	30.31
MMA+LA	97.60	93.57	96.81	92.20	**74.54**	71.79	86.85	79.57	91.00	88.16	34.40	28.47
Candle	95.84	92.46	95.89	91.77	73.49	69.70	84.92	76.26	90.75	87.51	38.02	32.63
LAMP without calibration	97.55	92.73	96.54	92.83	74.04	71.50	87.58	82.42	90.55	87.83	37.04	31.98
LAMP with calibration	**97.88**	**93.92**	**96.87**	**93.56**	74.46	**71.99**	**88.95**	**82.89**	**91.62**	**89.02**	**39.04**	**34.54**

Table 2 continued

Method | imb = 20	Caltech101		OxfordPets		Stanfordcars		Flower102		Food101		FGVC-Aircraft	
	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M
CoOp+LA	68.93	67.38	55.18	38.66	62.64	51.01	60.12	57.60	67.01	60.05	68.86	63.22
CoCoOp+LA	77.29	69.03	61.31	52.16	58.82	42.93	71.70	68.36	69.58	63.50	73.30	67.10
LFA	77.30	69.45	66.07	56.51	68.74	55.22	81.80	79.51	69.20	63.11	75.15	69.29
MaPLe+LA	78.80	70.02	67.47	57.50	83.28	73.42	83.25	**82.60**	69.95	63.80	78.48	72.76
TCP+LA	78.23	69.95	67.14	56.97	84.51	72.35	81.91	79.64	68.87	62.58	78.27	72.31
MMA+LA	76.75	67.28	67.64	58.22	83.79	73.27	80.76	78.28	68.03	61.97	78.02	72.07
Candle	78.53	70.11	67.32	57.23	80.96	71.01	82.59	81.51	67.50	60.80	77.80	71.91
LAMP without calibration	79.37	71.04	69.11	58.84	80.05	69.49	83.00	82.16	70.40	65.10	78.66	73.27
LAMP with calibration	**79.59**	**71.23**	**69.30**	**60.01**	**84.65**	**75.40**	**83.46**	82.23	**71.20**	**65.49**	**79.73**	**74.57**

We report the harmonic mean of accuracy (HM-A) and harmonic mean of macro-F1 (HM-M) on base and new classes. LAMP without calibration is the purely inductive variant directly comparable to inductive baselines, while LAMP with calibration additionally uses unlabeled new-class training images only for LBO bias calibration. The best results are marked with bold

Table 3 Results on long-tailed B2N benchmarks with an imbalance ratio of 50 averaged over 5 runs

Method | imb = 50	Caltech101		OxfordPets		Stanfordcars		Flower102		Food101		FGVC-Aircraft	
	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M
CoOp+LA	93.30	89.46	93.34	89.09	67.18	61.48	75.45	67.73	87.65	84.20	29.20	20.17
CoCoOp+LA	94.90	89.70	95.44	91.02	69.97	64.32	76.84	68.67	91.10	89.53	31.45	22.09
LFA	94.23	88.09	86.76	85.51	67.95	62.17	82.81	72.80	87.73	84.65	30.75	21.25
MaPLe+LA	95.57	90.70	**96.85**	91.80	71.96	66.60	85.06	78.10	89.70	86.05	33.40	25.89
TCP+LA	95.39	90.49	95.96	90.56	70.81	65.35	86.67	78.39	90.03	87.17	32.72	24.32
MMA+LA	96.43	91.23	96.89	91.85	72.48	66.84	85.79	77.40	89.05	85.63	35.98	27.21
Candle	94.95	90.40	95.83	91.33	71.78	66.30	84.62	76.03	90.70	87.76	36.68	28.13
LAMP without calibration	96.98	92.10	96.50	92.44	73.71	68.56	88.04	79.71	90.42	87.12	37.35	30.00
LAMP with calibration	**97.09**	**92.54**	96.35	**92.85**	**73.79**	**68.41**	**88.25**	**80.33**	**91.20**	**88.87**	**38.40**	**30.20**

Method | imb = 50	SUN397		DTD		EuroSAT		UCF101		ImageNet		✦ Average ✦	
	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M	HM-A	HM-M
CoOp+LA	65.91	60.78	51.42	36.11	57.35	50.29	61.62	53.66	65.59	59.59	68.00	61.14
CoCoOp+LA	76.18	69.97	59.37	44.76	64.99	58.91	77.53	68.53	68.70	61.34	73.32	66.26
LFA	75.13	69.05	61.78	47.99	61.91	55.15	79.49	71.79	68.21	61.01	72.43	65.41
MaPLe+LA	76.80	70.10	63.40	52.90	78.60	68.03	79.30	72.40	69.36	62.10	76.36	69.52
TCP+LA	77.20	69.44	66.78	56.95	81.87	71.09	79.93	72.08	68.05	60.79	76.86	69.69
MMA+LA	77.91	68.35	66.10	56.18	**83.41**	**74.37**	80.86	73.25	67.20	59.25	77.46	70.14
Candle	78.05	69.29	65.69	55.37	80.17	70.89	81.72	74.00	66.75	58.40	76.99	69.81
LAMP without calibration	78.24	69.65	66.08	58.85	79.96	68.87	82.06	80.55	69.97	64.24	78.12	72.01
LAMP with calibration	**78.50**	**70.22**	**66.83**	**59.35**	83.28	74.10	**82.50**	**78.78**	**70.45**	**64.67**	**78.79**	**72.76**

We report the harmonic mean of accuracy (HM-A) and harmonic mean of macro-F1 (HM-M) on base and new classes. LAMP without calibration is the purely inductive variant directly comparable to inductive baselines, while LAMP with calibration additionally uses unlabeled new-class training images only for LBO bias calibration. The best results are marked with bold

Table 4 Results on few-shot generalization benchmark averaged over 5 runs

Method	✦ Average ✦			ImageNet			Caltech101			OxfordPets			StanfordCars			Flowers102		
	Base	New	HM	Base	New	HM	Base	New	HM	Base	New	HM	Base	New	HM	Base	New	HM
CoOp	82.69	63.22	70.83	76.47	67.88	71.92	98.00	89.81	93.73	93.67	95.29	94.47	78.12	60.40	68.13	97.60	59.67	74.06
CoCoOp	80.47	71.69	75.44	75.98	70.43	73.10	97.96	93.81	95.84	95.20	97.69	96.43	70.49	73.59	72.01	94.87	71.75	81.71
MaPLe	82.28	75.14	78.24	76.66	70.54	73.47	97.74	94.36	96.02	95.43	97.76	96.18	72.94	74.00	73.47	95.92	72.46	82.56
TCP	83.20	76.94	79.72	77.31	71.00	74.02	**98.40**	94.00	96.15	95.40	**98.07**	**96.72**	78.50	73.10	75.70	97.77	75.93	85.48
MMA	84.13	75.36	79.26	77.27	69.87	73.38	98.23	94.67	**96.42**	94.67	97.20	95.92	80.80	74.13	77.32	97.73	75.57	85.23
Candle	83.88	76.53	79.87	76.97	68.54	72.48	98.54	94.47	96.46	95.53	97.34	96.43	79.14	74.92	76.97	98.01	77.52	86.57
LAMP	**85.23**	**77.18**	**80.83**	**77.62**	**71.60**	**74.49**	98.08	**94.02**	96.01	**95.65**	97.82	**96.72**	**81.09**	**75.40**	**78.14**	**98.04**	**77.98**	**86.87**
Method	Food101			FGVCAircraft			SUN397			DTD			EuroSAT			UCF101		
	Base	New	HM	Base	New	HM	Base	New	HM	Base	New	HM	Base	New	HM	Base	New	HM
CoOp	88.33	82.26	85.19	40.44	22.30	28.75	80.60	65.89	72.51	79.44	41.18	54.24	92.19	54.74	68.69	84.69	56.05	67.45
CoCoOp	90.70	91.29	90.99	33.41	23.71	27.74	79.74	76.86	78.27	77.01	56.00	64.85	87.49	60.04	71.21	82.33	73.45	77.64
MaPLE	90.71	92.05	**91.38**	37.44	35.61	36.50	80.82	78.70	79.75	80.36	59.18	68.16	94.07	73.23	82.35	83.00	78.66	80.77
MMA	90.13	91.30	90.71	40.57	36.33	38.33	82.27	78.57	80.38	83.20	**65.63**	73.38	85.46	**82.34**	83.87	86.23	80.03	83.01
TCP	**90.57**	**91.37**	90.97	41.97	34.43	37.83	82.63	78.20	80.35	82.77	58.07	68.25	91.63	74.73	82.32	87.13	**80.77**	83.83
Candle	90.52	91.23	90.87	43.86	**36.69**	39.96	81.64	77.93	79.74	81.40	61.35	69.97	89.97	81.33	85.43	87.13	80.51	83.69
LAMP	90.40	91.03	90.71	**45.95**	36.66	**40.78**	**82.92**	**79.00**	**80.91**	**85.04**	65.26	**73.85**	**93.90**	80.42	**86.63**	**88.79**	79.83	**84.07**

We report the accuracy of Base / New classes and their harmonic mean (HM). The best results are marked with bold

Table 5 Results of Long-tailed Cross-Dataset Generalization benchmark averaged over 5 runs

Method	ImageNet ***(Source)***		Caltech101		Oxfordflowers		Stanfordcars		Flower102		Food101	
	Acc	Mac	Acc	Mac	Acc	Mac	Acc	Mac	Acc	Mac	Acc	Mac
CoOp+LA	70.75	63.92	90.85	88.78	86.85	82.93	64.95	61.67	67.12	59.41	85.02	83.85
CoCoOp+LA	71.24	64.40	91.31	89.39	88.76	84.14	65.52	63.00	69.50	61.50	86.30	84.12
MaPLe+LA	70.18	63.35	90.17	88.02	89.29	85.49	65.09	62.43	70.38	61.87	86.05	84.00
TCP+LA	71.03	64.34	90.56	88.10	89.37	85.66	64.78	62.36	71.15	62.26	86.45	84.17
MMA+LA	70.65	63.98	90.21	87.51	88.29	84.38	63.87	62.01	71.10	62.23	85.80	83.81
Candle	70.92	64.62	91.40	89.78	88.96	85.82	64.60	62.57	68.94	58.51	85.39	83.25
LAMP	**72.70**	**66.28**	**91.85**	**90.14**	**90.12**	**87.07**	**65.48**	**62.98**	**72.26**	**63.40**	**86.58**	**84.31**

Method	FGVC-Aircraft		SUN397		DTD		EuroSAT		UCF101		Avg. ***(w/o Source)***	
	Acc	Mac	.Acc	Mac	Acc	Mac	Acc	Mac	Acc	Mac	Acc	Mac
CoOp+LA	18.92	12.00	63.25	59.92	42.46	32.71	44.40	40.69	65.98	64.70	63.69	59.14
CoCoOp+LA	22.80	14.82	66.07	61.73	45.04	35.64	42.86	38.34	67.54	66.45	65.18	60.32
MaPLe+LA	23.95	14.49	66.48	61.96	45.47	35.68	45.29	41.01	67.01	66.07	65.40	60.40
TCP+LA	22.03	14.08	**66.85**	61.64	49.20	38.48	48.89	43.12	67.28	66.40	66.14	60.96
MMA+LA	25.02	16.52	65.98	60.26	49.50	39.10	48.65	42.99	65.57	64.35	65.88	60.65
Candle	24.18	15.35	66.17	60.43	48.10	37.96	48.49	42.26	67.20	66.15	65.85	60.61
LAMP	**26.24**	**17.83**	66.73	**62.10**	**50.47**	**40.62**	**50.40**	**44.64**	**68.97**	**67.64**	**67.44**	**62.46**

All models are trained on long-tailed ImageNet with $\texttt{imb} = 100$, then evaluated over another target dataset. We report the accuracy and macro-F1. The best results are marked with bold

Table 6 Results of Long-tailed domain generalization benchmark averaged over 5 runs

Method	ImageNet ***(Source)***		ImageNet-A		ImageNet-V2		ImageNet-S		ImageNet-R	
	Acc	Mac	Acc	Mac	Acc	Mac	Acc	Mac	Acc	Mac
CoOp+LA	70.75	63.92	48.55	45.94	63.53	60.12	47.24	45.01	73.67	64.05
CoCoOp+LA	71.24	64.40	48.87	46.00	63.62	60.44	47.89	45.83	74.45	64.94
MaPLe+LA	70.18	63.35	49.32	45.32	63.70	60.45	48.05	45.69	74.68	65.02
TCP+LA	71.03	64.34	48.79	45.91	63.24	59.75	48.00	45.68	74.93	65.31
MMA+LA	70.65	63.98	50.12	46.12	63.55	59.97	47.65	45.32	75.34	65.96
Candle	70.92	64.62	49.16	45.28	62.70	59.46	48.38	45.94	75.20	65.80
LAMP	**72.70**	**66.28**	**52.92**	**48.30**	**65.51**	**61.67**	**49.20**	**48.79**	**76.46**	**67.47**

All methods are trained on an imbalanced ImageNet with $\texttt{imb} = 100$ and directly evaluated on other four target datasets. We report the accuracy and macro-F1. The best results are marked in bold

these diverse and demanding contexts provides strong evidence of its enhanced generalization capability. This ability to adapt effectively to varied data distributions highlights a key strength of the framework, suggesting its practical utility for a wide range of real-world applications.

4.2.4 Results of LT-DGB

The experimental results are summarized in Table 6. LAMP improves both the source-domain results and the out-of-distribution results, showing that the learned prompts retain transferable visual-language knowledge under long-tailed source training. This performance gain strongly indicates that LAMP is more effective and inherently adept at adapting to and learning from the long-tailed distributions characteristic of the source training sets. Furthermore, when evaluated on the four distinct component datasets derived from ImageNet, LAMP continues to demonstrate clear superiority. This robust performance under varied target conditions convincingly demonstrates the strong and generalized ability of LAMP to transfer knowledge effectively to different visual domains.

4.2.5 Efficiency Study

We further compare the complexity of LAMP with CoOp/MaPLe in Table 7. Detailed runtime statistics were measured on an NVIDIA H20 GPU with ViT-B/16. The top-K retrieval over

Table 7 Detailed efficiency comparison on ViT-B/16 with imb = 10

Method	Trainable Param	Tra. Time	HM-A	HM-M
CoOp	8K (+0.005%)	0.5s/Batch	69.23	64.50
MaPLe	4.26M (+2.85%)	2.0s/Batch	78.56	73.31
LAMP	**354.7K (+0.24%)**	**1.6s**/Batch	80.93	75.66

Bold values indicate the best performance among the compared methods

Table 8 Ablation of retrieved prompt modalities in MPP

Retrieved prompts	HM-A	HM-M
Visual prompts only	80.69	76.25
Text prompts only	81.19	76.69
Visual + text prompts	**81.24**	**76.95**

The results are averaged over imb = 10 and imb = 20
Bold values indicate the best performance among the compared methods

MPP introduces only a small search cost over M prompts at the selected layers, i.e., $O(|\mathtt{Sec}|MD)$ per instance, which is negligible compared to the transformer blocks of CLIP. More importantly, this retrieval does not require an additional CLIP forward pass, since it merely selects a small subset of prompt tokens before the standard transformer computation. As shown in Table 7, LAMP uses only 354.7K trainable parameters, accounting for 0.24% of CLIP parameters and roughly 12 times fewer than MaPLe. LAMP also reduces the per-batch time from 2.0s to 1.6s with a batch size of 32, while improving HM-A by 2.37 and HM-M by 2.35. These results indicate that the dynamic prompt pool does not sacrifice efficiency for accuracy. The primary runtime bottleneck remains the frozen CLIP backbone, and the smaller trainable prompt space of LAMP offsets the additional retrieval cost. The main computational overhead of LAMP lies in optimizing the prompts and projectors, similar to previous state-of-the-art methods. However, LAMP requires fewer tunable parameters because the prompt dimension is smaller than that of the visual and textual hidden embeddings, as confirmed by the ablation studies. Beyond these efficiency gains, LAMP achieves substantial performance improvements, validating its effectiveness.

4.3 Ablation Studies

In this subsection we perform ablation studies on key components and hyper-parameters of LAMP.

4.3.1 Different Components

Recall that LAMP comprises three essential components: the Multi-modal Prompt Pool (MPP), Modality-Shared Prompts (MSP), and Load Balancing Optimization (LBO). We systematically evaluate the contribution of each component and their combinations through an ablation study, with results reported in Table 10. The ablation is conducted on the long-tailed B2N benchmark using ImageNet and Food101.

Our analysis shows that employing only MPP and MSP yields an aggregate improvement of approximately 2.5% in both accuracy and macro-F1 over the baseline CLIP model. Notably, MSP contributes slightly more to this gain than MPP, which we attribute to the inherent design of MSP, aimed at capturing synergistic cross-modal semantics. The effect of LBO becomes more pronounced when combined with the prompt pool. On ImageNet, MPP+LBO surpasses MPP by 2.37 HM-A and 1.75 HM-M; on Food101, the corresponding improvements are 1.19 and 1.34. These gains are larger than what would be expected from merely adding a small number of prompt parameters, indicating that LBO changes how samples are assigned to prompt experts, thereby preventing head-class samples from dominating a narrow subset of prompts throughout training. Adding MSP further improves the MPP+LBO combination, suggesting that load balancing and cross-modal sharing are complementary rather than redundant. The complete integration of all three components (MPP+MSP+LBO) achieves the highest performance, confirming that MPP, MSP, and LBO work synergistically to enhance model capability.

We further isolate two design choices specific to MPP and LBO: we compare visual-prompt-only injection, text-prompt-only injection, and the default visual-text prompt injection in Table 8 under the same visual-query retrieval mechanism, and we evaluate the calibrated LBO setting against a purely inductive variant without new-class calibration in Table 9.

Table 8 shows the distinct roles of the retrieved visual and text prompts after the same visual-query retrieval step. Using only text prompts slightly outperforms using only visual prompts, achieving 81.19 HM-A and 76.69 HM-M versus 80.69 HM-A and 76.25 HM-M. This suggests that text prompts provide a more stable semantic anchor for class-level alignment, which is particularly beneficial for HM-M since tail-class errors are more pronounced in macro-F1. Visual prompts remain necessary, as they adapt the image branch at the instance level and supply visual evidence for the retrieved prompt pair. When both visual and text prompts are injected, the model attains the best HM-A and HM-M. Although the gain over the text-prompt-only setting is moderate, it is consistent with the design of MPP: text prompts primarily strengthen semantic discrimination, while visual prompts help align the image representations with these semantics. Therefore, the final design uses a visual query to retrieve paired visual–text prompts and injects both prompt modalities, rather than treating MPP as a text-only or visual-only prompt pool.

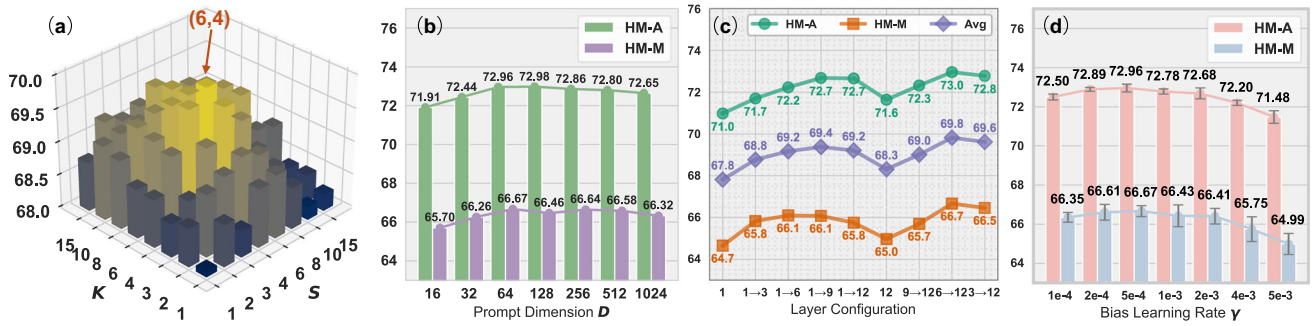


Fig. 3 The ablation studies of **a** K and S; **b** Prompt dimension D; **c** layer configuration; **d** bias (initial) learning rate, with standard deviations. All ablation studies were conducted on ImageNet

4.3.2 LBO Calibration

Table 9 further separates the contribution of LBO from that of bias calibration using unlabeled new-class images on a representative subset. This calibrated setting corresponds to the LAMP with calibration rows in the main comparison tables, while the LAMP without calibration rows provide the direct inductive comparison. Across all listed datasets and imbalance ratios, calibration improves the average HM-A from 79.22 to 80.62 and the average HM-M from 73.49 to 75.35. The magnitude of the gains varies across datasets: they are modest on ImageNet, Caltech101, and OxfordPets, where prompt assignments are already relatively stable, but become more pronounced on FGVC-Aircraft and EuroSAT. On FGVC-Aircraft, HM-M increases by 2.38 and 2.56 under imbalance ratios 10 and 20, indicating improved handling of fine-grained tail categories. On EuroSAT, HM-M increases by 5.63 and 5.91, suggesting that calibration is beneficial when scene-level categories exhibit high visual diversity and prompt usage is sensitive to the sampled tail images. These findings align with the objective of LBO: after prompt-load calibration, visually diverse tail classes are less likely to be absorbed by a few overused head-class prompts. Thus, LBO improves the overall balance of prompt usage, while the purely inductive variant remains competitive and preserves the fair comparison protocol.

4.3.3 Number of Retrieved Prompts K and Length of Shared Prompts S

We present a grid search over these two key hyper-parameters on the long-tailed B2N benchmark in Fig. 3.a, using the average of HM-A and HM-M as the performance metric. Our analysis shows that setting both hyper-parameters too low leads to a more pronounced performance degradation. Conversely, we observe a slight but consistent drop in performance when K and S become too large (typically at or above 8), a trend that may be attributed to overfitting on the training set. Based on these results, we choose $K = 6$ and $S = 4$ as the optimal configuration.

4.3.4 Prompt Dimension

The prompt dimension intuitively determines the upper bound of model adaptation capacity during fine-tuning. We present a detailed ablation study on the long-tailed B2N benchmark in Fig. 3b. The results show that both primary evaluation metrics improve considerably as the dimension increases from 16, but their gains level off markedly beyond dimension 64, with performance exhibiting only minor fluctuations at higher values. This saturation indicates diminishing returns in model expressiveness. Meanwhile, higher-dimensional prompts incur substantially increased computational and memory costs. Therefore, 64 is the optimal dimension on this ablation curve; all main comparison tables use a prompt dimension of 64 under the unified protocol to maintain a consistent computational budget across datasets.

4.3.5 Layers for Adding Prompts

We conduct an ablation study on this factor, with results reported on the long-tailed B2N benchmark in Fig. 3.c. The results show a clear dependence on the selected layer range. Using only the first layer yields an average score of 67.8, and extending the range gradually from layers 1–3 to 1–9 improves the average score from 68.8 to 69.4. However, inserting prompts into all layers 1–12 does not bring further improvement; instead, the average score drops to 69.2. Using only the last layer is also insufficient, giving 68.3. The best configuration is layers 6–12, achieving 73.0 HM-A, 66.7 HM-M, and an average score of 69.8. This pattern indicates that LAMP benefits most from mid-to-deep layers, where the representations are semantic enough for prompt experts to guide class-level clustering, while shallow layers are more sensitive to local visual patterns. We therefore adopt layers 6–

Table 9 Effect of LBO calibration on unlabeled new-class training images

Setting	Metric	Avg.	Caltech101		OxfordPets		Food-101		FGVC-Aircraft		EuroSAT		ImageNet	
			10	20	10	20	10	20	10	20	10	20	10	20
with calibration	HM-A	**80.62**	**97.58**	**97.88**	**97.76**	**96.87**	**92.45**	**91.62**	**40.69**	**39.04**	**84.78**	**84.65**	**72.96**	**71.20**
	HM-M	**75.35**	**92.91**	**93.92**	**93.09**	**93.56**	**91.15**	**89.02**	**32.13**	**34.54**	**76.36**	**75.40**	**66.67**	**65.49**
without calibration	HM-A	79.22	96.86	97.55	97.30	96.54	91.67	90.55	38.45	37.04	82.07	80.05	72.18	70.40
	HM-M	73.49	92.50	92.73	92.69	92.83	90.22	87.83	29.75	31.98	70.73	69.49	65.99	65.10

HM-A and HM-M are reported as mean results. The Avg. column is computed over all listed datasets and imbalance ratios
Bold values indicate the best performance among the compared methods

Table 10 Ablation studies of components, evaluated with imb = 10 on two datasets

component	ImageNet		Food101	
	HM-A	HM-M	HM-A	HM-M
CLIP	68.05	62.45	89.25	88.83
MPP	70.05	64.08	90.87	89.50
MSP	70.94	64.64	91.21	89.54
MPP+MSP	71.23	64.70	91.10	89.87
MPP+LBO	72.42	65.83	92.06	90.84
MPP+MSP+LBO	72.96	66.67	92.45	91.15

Table 11 Ablation studies of different β

β	HM-A	HM-M	Avg
0.00	71.95	65.70	68.83
0.20	71.49	66.02	69.25
0.50	72.67	66.50	69.59
1.00	72.79	66.58	69.69
2.00	72.96	66.67	69.82
4.00	72.89	66.04	69.47
8.00	72.84	65.96	69.40

12 as the default in our main experiments, which provides the strongest result without inserting modules into every layer.

4.3.6 Initial Bias Learning Rate γ

As noted earlier, we employ a cosine scheduler to dynamically adjust γ and focus our ablation on its initial value. The corresponding results are shown in Fig. 3.d. Our analysis reveals a severe performance drop when the initial γ exceeds $1e-3$, accompanied by considerable instability in the HM-M metric. This suggests that larger initial values of γ tend to introduce instability into the LBO process. Our experiments indicate that setting the initial value to $\gamma = 5e-4$ generally yields the best results.

Table 12 Ablation Studies of M

M	10	12	15	20	25	30
HM-A	72.28	72.4	72.87	72.96	73	72.96
HM-M	65.96	66.12	66.5	66.67	66.65	66.48

4.3.7 Balance Coefficient β

We present the ablation results for the similarity loss coefficient β in Table 11. Our key finding is that omitting the similarity loss entirely ($\beta = 0$) leads to a statistically significant performance drop of approximately 1% on average across all evaluated metrics, highlighting the contribution of this loss term. We attribute this improvement to the auxiliary loss, which facilitates more effective gradient rebalancing across the model's parameter space. Specifically, it mitigates the sharp fluctuations of LBO that occur during the initial, unstable phase of training, thereby promoting smoother convergence. Table 11 shows that $\beta = 2$ yields slightly higher scores than $\beta = 1$ in isolation; we keep $\beta = 2$ in all main comparisons to maintain consistency with the unified protocol.

4.3.8 Prompt Pool Size M

We further examine the impact of M by fixing $K = 6$ and varying M from 10 to 30, following (Dong et al., 2022a). The results on ImageNet under the long-tailed B2N benchmark are reported in Table 12. We find that increasing M beyond 20 yields only a marginal improvement (0.04%), likely because M has little effect on the number of prompts actually activated. As a trade-off between performance and efficiency, we set $M = 20$.

4.3.9 Stability Analysis

We further evaluate the stability of LAMP on small-scale and fine-grained datasets, where prompt retrieval can be more sensitive to sample scarcity. Table 13 reports the mean results and standard deviations over 10 seeds on FGVC-Aircraft and

Table 13 Stability analysis on small datasets

Dataset	imb	HM-A	HM-M
FGVC-Aircraft	10	40.75±0.91	32.15±2.40
	20	39.09±1.73	34.58±2.78
EuroSAT	10	83.54±7.70	74.56±7.57
	20	80.53±6.11	71.15±5.89

Results are reported as mean ± standard deviation over 10 seeds

EuroSAT; these statistics are also included along with other supplementary standard deviations, in Appendix Table 19.

The results reveal two distinct sources of difficulty. FGVC-Aircraft involves subtle inter-class differences, so its absolute HM-M is lower than that on general object datasets. Nevertheless, performance does not collapse when the imbalance ratio increases from 10 to 20; HM-M even improves from 32.15 to 34.58. This indicates that the retrieved prompts still manage to capture fine-grained cues for rare classes, even though the class boundaries remain challenging. EuroSAT attains much higher HM-A and HM-M but shows larger standard deviations, suggesting that the satellite domain is sensitive to the particular tail samples selected in each run. Therefore, the stability issue is not an inherent weakness of MPP retrieval; it depends on whether the primary challenge is fine-grained discrimination, as in FGVC-Aircraft, or high intra-class visual diversity under limited tail samples, as in EuroSAT.

4.3.10 Feature Visualization

To better illustrate how LAMP reshapes the learned feature geometry, we visualize the image features using t-SNE in Fig. 4. Compared with the baseline prompt tuning result, LAMP produces more compact class-wise clusters and clearer separation among neighboring classes. This visualization supports the motivation behind MPP and LBO, where dynamic prompt retrieval encourages samples with similar semantics to share appropriate prompt experts, and load balancing prevents excessive concentration on head-class prompts and promotes a more balanced feature organization.

5 Case Study

To further justify the theoretical expressivity of LAMP, in this section we aim to study an example that vanilla prompt learning (VPL) fails to solve but LAMP can solve. Inspired by Wang et al. (2023), Hu et al. (2024), we raise the following exemplar for discussion:

Case 1 **An exemplar model**: a pretrained single-layer single-head transformer T consisting of an Attention layer $Attn$ and a MLP layer MLP:

$$MLP(x) = \begin{cases} (W_2W_1 + I)x, & [W_1x]_0 > 0 \\ x, & [W_1x]_0 \le 0 \end{cases} \tag{12}$$

An exemplar dataset: $\{(X_1 = [x_1, x_0], Y_1 = [y_1, y_0]), (X_2 = [x_2, x_0], Y_2 = [y_{21}, y_{20}])\}$, where $Y_1 \in C^{base}$ and $Y_2 \in C^{new}$.

Theorem 2 *For VPL, there exists no $P \in R^{n_p \times d}$ that simultaneously satisfies $Y_0 = T([P, X_0])$ and $Y_1 = T([P, X_1])$; However, for LAMP, this is conditionally satisfiable.*

Proof Some of our proof follows (Wang et al., 2023) but expands their conclusions. Generally, we first define the convex cone of any vector combination $\{\mathbf{v}_0, \mathbf{v}_1 ... \mathbf{v}_n\}$ as $\text{Cone}(\mathbf{v}_0, \mathbf{v}_1 ... \mathbf{v}_n) = \{x : \sum_{i=1}^{m} v_i \mathbf{v}_i, v_i > 0\}$. Considering that W_2W_1 is a non-singular matrix, then we can find at most two points for the inverse projection for $MLP(x_0)$ from $\mathcal{Y}$ to $\mathcal{X}$, i.e $MLP^{-1}(y) = \{y, (W_2W_1 + I)^{-1}y\}$. We rename them as y_{10} and y_{20}.

Then, we can find $\mathbf{c}_1$ and $\mathbf{c}_2$, satisfying $\mathbf{c}_1, \mathbf{c}_2 \perp y_{10} - x_0$, $\mathbf{c}_1, \mathbf{c}_2 \perp y_{20} - x_0$, $(W_2W_1 + I)^{-1}y_{10} - x_0$, $(W_2W_1 + I)^{-1}y_{20} - x_0$, and $c_1 \perp c_2$. We can pick up x_1 and x_2 such that $Cone(-Attn(x_0, X_1), a - x_0) \cap Cone(-Attn(x_0, X_2), b - x_0) = \emptyset$, subject to $\forall a \in \{y_{10}, (W_2W_1 + I)^{-1}y_{10}\}$ and $\forall b \in \{y_{20}, (W_2W_1 + I)^{-1}y_{20}\}$. Suppose, for contradiction, that there exists P that satisfies both $Y_0 = T([P, X_0])$ and $Y_1 = T([P, X_1])$, i.e.:

$$\begin{aligned} &Attn(x_0, [P, X_1]]) = \lambda(X_1, x_0, [P, X_1])Attn(x_0, X_1) \\ &+ \lambda(P, x_0, [P, X_1])Attn(x_0, P), \end{aligned} \tag{13}$$

The same equation for X_2 is abbreviated. Therefore, we can obtain the following formula:

$$\begin{aligned} &Attn(q_i, [P, X_0]) \\ &= \frac{q_i \sum_{j \in I_P} ([P, X_0]_j^T)^2}{q_i \sum_j [P, X_0]_j} \\ &\quad + \frac{q_i \sum_{j \in I_{X_0}} ([P, X_0]_j^T)^2}{q_i \sum_j [P, X_0]_j} \\ &= \frac{q_i \sum_{j \in I_P} [P, X_0]_j}{q_i \sum_j [P, X_0]_j} \frac{q_i \sum_{j \in I_P} ([P, X_0]_j^T)^2}{q_i \sum_{j \in I_P} [P, X_0]_j} \\ &\quad + \frac{q_i \sum_{j \in I_{X_0}} [P, X_0]_j}{q_i \sum_j [P, X_0]_j} \frac{q_i \sum_{j \in I_{X_0}} ([P, X_0]_j^T)^2}{q_i \sum_{j \in I_{X_0}} [P, X_0]_j} \\ &= \alpha \cdot Attn(q_0, P) + (1 - \alpha) \cdot Attn(q_0, X_0) \end{aligned} \tag{14}$$

I_P and I_{X_0} represents their index set respectively. α is the coefficient controlling the balance between two types of

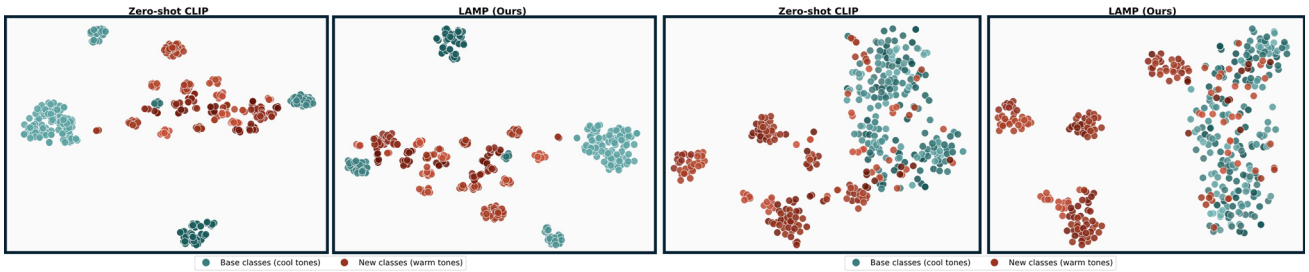


Fig. 4 T-SNE visualization of image features under the long-tailed setting. LAMP yields more compact and better separated feature clusters, as well as more distinctness between base and new classes

attention. Xu et al. (2023) provides in-depth discussions to this term, and finds that prompt tuning tends to learn less similar prompts for adaptation, but under the risk of discarding the pre-trained knowledge. Returning to our derivation, the following also holds true:

$$\begin{aligned} &Attn(x_0, [P, X_1]) + x_0 \in MLP^{-1}(y_{10}), \\ &Attn(x_0, [P, X_2]) + x_0 \in MLP^{-1}(y_{20}) \end{aligned} \tag{15}$$

These results imply that $Attn(x_0, P)$ must be in both $Cone(a - x_0, -Att(x_0, X_1))$ and $Cone(b - x_0, -Att(x_0, X_2))$ where $a \in \{y_{10}, (W_2W_1 + I)^{-1}y_{10}\}$ and $b \in \{y_{20}, (W_2W_1 + I)^{-1}y_{20}\}$. This contradicts the aforementioned hypothesis that the cones are distinct, meaning that prompt learning fails on this exemplar. We then show how LAMP solves this case. Starting from 14, we have:

$$\begin{aligned} &A(x_0, [P^*_{y_0}, X_1]]) = \lambda(X_1, x_0, [P^*_{y_0}, X_1]) \cdot \\ &A(x_0, X_1) + \lambda(P^*_{y_0}, x_0, [P^*_{y_0}, X_1]) A(x_0, P^*_{y_0}) \\ &A(x_0, [P^*_{y_1}, X_2]]) = \lambda(X_1, x_0, [P^*_{y_1}, X_2]) \cdot \\ &A(x_0, X_2) + \lambda(P^*_{y_1}, x_0, [P^*_{y_1}, X_2]) A(x_0, P^*_{y_1}) \end{aligned} \tag{16}$$

where $Attn$ is abbreviated as A, $P^*_{y_c}$ stands for the collection of $\{P_i\}, \{P^v_i\}, \{P^t_{i,c}\}$. Similarly, we can derive the following:

$$\begin{aligned} &Attn(x_0, [P^*_{y_0}, X_1]) + x_0 \in MLP^{-1}(y_{10}), \\ &Attn(x_0, [P^*_{y_1}, X_2]) + x_0 \in MLP^{-1}(y_{20}) \end{aligned} \tag{17}$$

Accordingly we can assume that $Attn(x_0, P^*_{y_0})$ is in $Cone(a - x_0, -Att(x_0, X_1))$ while $Attn(x_0, P^*_{y_1})$ is in $Cone(b - x_0, -Att(x_0, X_2))$. By fixing the shared prompts and only tuning the extra visual or textual prompts, we can find corresponding optimal $P^{v+\dagger}_i$ and $P^{t+\dagger}_i$ such that the above requirements can be satisfied. However, if the extra prompts defined by MPP and MSP are discarded from LAMP, the theoretical expressivity upper bound for LAMP degrades to that of the vanilla prompt methods. □

6 Conclusion

While most existing multi-modal prompt learning methods are designed for the balanced dataset, real-world datasets always follow a long-tailed distribution, stressing the importance of designing prompting methods over imbalanced data. LAMP introduces three crafted and co-designed mechanisms: multi-modal prompt pool, modality-shared prompts and load balancing optimization. Extensive comparisons and ablation studies validate the superiority and effectiveness of LAMP.

Appendix

LAMP: Long-Tailed Multi-modal Prompt Learning

The content of the Appendix is summarized as follows:

- in Sect. A, we introduce datasets and tasks.
- in Sect. B, we introduce methods for comparison.
- in Sect. C, we introduce extra ablation studies.
- in Sect. D, we give more experiments.
- in Sect. E, we provide statement of usage of LLM.
- in Sect. F, we provide reproducibility statement.

Appendix A: Details of Datasets and Tasks

Detailed introduction of datasets are listed here:

- DTD (Describable Texture Dataset) is a texture datasets that containing 5640 images with the size ranging from 300x300 to 640x640. The dataset has 47 classes.
- Food-101 contains 101,000 images concerning different types of food, with a uniform size of 512x512. These images are from 101 classes, where each class consists of 750 images for training and 250 for testing.
- UCF-101 is a action recognition dataset containing 13,320 videos from 101 action classes. The total length

of videos is 27 h. For multimodal learning, these videos are sampled into 11422 images, with 7639 for training and 3783 for testing.

- SUN397 is a scene understanding dataset containing 108,754 images from 397 scene classes. Each class has as least 100 images.
- EuroSAT is a land-use 10-class classification dataset based on Sentinel-2 satellite images. It has 27,000 images with the size of 64x64. Each class has about 2000-3000 images.
- FGVC-Aircraft is a aircraft dataset containing 10200 hih-resolution images, including 102 classes with each contains 100 images.
- StanfordCars is a fine-grained car classification dataset, containing 16185 images from 196 classes. The training set contains 196 classes, and the number of training and tests 8144 and 8041.
- Caltech-101 is a classical object recognition dataset, containing 101 classes and 9000 images. The classes are imbalancedly distributed.
- OxfordFlowers contains 102 classes of flowers that commonly seen in British, with 40-258 classes in each class.
- OxfordPets contains 7349 images from 37 classes, with each image annotated with the segmentation and class label.
- ImageNet(1K) is a subset of ImageNet full dataset and contains 1000 classes. There are a total of 1,281,167 training images, 50,000 validation images and 100,000 testing images.
- ImageNet-V2 is an extended version of ImageNet, aiming to deal with the overfitting issue of original experiments. It includes three test sets: "Threshold0.7", "MatchedFrequency" and "TopImages", which are sampled from different strategies.
- ImageNet-R (ImageNet-Rendition) is a special test set for renditions. It includes renditions of 200 ImageNet classes, resulting in 30,000 images.
- ImageNet-A (ImageNet-Adversarial) is proposed for discovering the robustness of the model to adversarial attacks. These modifications for attack are nearly invisible to human but would result in misclassification.
- ImageNet-S (ImageNet-Sketch) is a test set for sketch arts. It contains sketch images for the same classes as original ImageNet.

We also list the prompts used for these datasets in Table 14.

Table 14 Prompts used for all the datasets

Datasets	Prompts
ImageNet	"a photo of a [CLASS]."
Caltech101	"a photo of a [CLASS]."
OxfordPets	"a photo of a [CLASS], a type of pet."
StanfordCars	"a photo of a [CLASS]."
Flowers102	"a photo of a [CLASS], a type of flower."
Food101	"a photo of a [CLASS], a type of food."
FGVCAircraft	"a photo of a [CLASS],a type of aircraft."
SUN397	"a photo of a [CLASS]."
DTD	"[CLASS] texture."
EuroSAT	"a centered satellite photo of [CLASS]."
UCF101	"a photo of a person doing [CLASS]."
ImageNet	"a photo of a [CLASS]."
ImageNet-V2	"a photo of a [CLASS]."
ImageNet-S(Sketch)	"a photo of a [CLASS]."
ImageNet-A	"a photo of a [CLASS]."
ImageNet-R	"a photo of a [CLASS]."

Appendix B Details of Methods for Comparison

- CoOp: CoOp (Zhou et al., 2022b) first introduces the prompt engineering in Natural Language Processing to Multimodal models. It adds extra prompts to the original input text to learn the optimal textual templates for downstream datasets.
- CoCoOp: CoCoOp (Zhou et al., 2022a) improves CoOp by constructing a vision-conditional textual prompt. The outputs of vision encoder are used to construct the textual prompts, thus achieving multi-modal interactions.
- MaPLe: MaPLe (Khattak et al., 2023a) introduces a layer-wise prompt learning technique, where each layer is plugged in a textual prompt, and the textual prompt is also used to generate a layer-wise visual prompt.
- TCP: TCP (Yao et al., 2024) proposes to incorporate final class embeddings, to further enhance the discriminability of prompts. Specifically, it introduces a Textual Knowledge Embedding (TKE) module to transfer the final class embeddings to the same space of intermediate textual tokens.
- MMA: MMA (Yang et al., 2024) introduces a modality-part-shared adapter between two encoders. Concretely, intermediate visual and textual tokens are first projected to a joint space, then processed with a modality shared MLP, and transformed back to their original space.
- (Extra) LA: Logit Adjustment Loss (LA) (Menon et al., 2020) is a tailored cross-entropy loss for imbalanced learning, where each term in the softmax function is appended with an extra term denoting the class cardinality of the corresponding class.

See Tables 15, 16 and 17.

Table 15 Results of different prompt tuning methods over 10 datasets under the imbalance ratios of 100

	Cal	Ox	SC	FLw	Food	FA	SUN	DTD	ES	UDF	Avg
MaPLe+LA	95.30	93.57	70.65	84.25	88.47	34.08	76.78	63.27	78.1	78.56	76.30
TCP+LA	94.47	92.90	71.76	85.4	89.67	32.70	77.00	65.49	80.3	79.03	76.87
MMA+LA	**96.97**	**93.94**	**73.79**	84.74	88.98	33.26	**76.69**	66.08	83.30	79.07	77.68
Candle	95.09	93.71	72.60	83.27	88.06	37.90	75.34	65.76	80.06	81.59	77.33
LAMP	95.75	93.00	73.13	**85.36**	**90.21**	**38.47**	76.22	**66.70**	**83.54**	**81.98**	**78.43**

Following Shi et al. (2024a), the models are trained over an imbalanced training set and inference over both base and new classes; we report the harmonic mean of testing accuracy over base and new classes. The best results are marked with bold

Table 16 Results of different prompt tuning methods over 10 datasets under the imbalance ratios of 200

	Cal	Ox	SC	FLw	Food	FA	SUN	DTD	ES	UDF	Avg
MaPLe+LA	92.82	94.29	69.73	82.06	88	31.29	75.84	60.47	76.76	75.82	74.70
TCP+LA	**94.01**	93.05	70.46	**84.77**	**89.84**	32.64	76.5	62.56	79.74	78.48	76.20
MMA+LA	90.56	94.81	70.09	81.35	88.4	29.08	77.13	60.04	82.09	77.74	75.12
Candle	92.63	92.89	71.4	82.86	86.01	35.29	75.08	**63.34**	79.17	79.3	75.79
LAMP	93.08	**94.87**	**71.98**	82.35	89.26	**37.3**	**75.87**	**63.34**	**79.85**	**80.35**	**76.82**

Following Shi et al. (2024a), the models are trained over an imbalanced training set and inference over both base and new classes; we report the harmonic mean of testing accuracy over base and new classes
Bold values indicate the best performance among the compared methods

Table 17 Results of M

M	HM-A	HM-M
10	72.28	65.96
12	72.40	66.12
15	72.87	66.50
20	72.96	66.67
25	73.00	66.65
30	72.96	66.48

Appendix C Further Ablation Studies

The impact of *M*

We further discuss the impact of M in this part. By fixing K as 6, we follow Dong et al. (2022a) to alter M from 10 to 30. Results conducted on ImageNet with Long-tailed B2N benchmark are shown in Table 17. We find that the performance gain by increasing M over 20 is quite minor (0.04%), which may result from the fact that M does not effectively influence the actual number of activated prompts; as a trade-off between performance and efficiency, we set M to 20.

Standard deviations of supplementary results

We provide the standard deviations of the main B2N results in Table 18. We further report the standard deviations of supplementary ablations in Table 19 and Table 20. These results complement the mean values reported in the main text. The variance pattern is also informative. On datasets such as Caltech101, Food-101, and ImageNet, the standard deviations are small across imbalance ratios, which indicates that the gains of LAMP are not caused by a favorable random seed. Larger deviations appear mainly on EuroSAT and FGVC-Aircraft under stronger imbalance. These two datasets correspond to the regimes discussed in the main text: EuroSAT is sensitive to the sampled tail images because scene-level satellite patterns are visually diverse, while FGVC-Aircraft is sensitive to fine-grained class boundaries. The standard deviations therefore help identify where long-tailed prompt tuning is intrinsically harder, rather than weakening the average trend.

Table 18 Standard deviations of LAMP on long-tailed B2N benchmarks (representative subset)

Metric	imb	Caltech101	OxfordPets	Food-101	FGVC-Aircraft	EuroSAT	UCF101	ImageNet
HM-A	10	97.58±0.38	97.76±1.51	92.45±0.61	40.69±0.87	84.78±7.81	87.70±1.28	72.96±0.18
	20	97.88±0.36	96.87±1.04	91.62±0.66	39.04±1.93	84.65±12.13	83.46±1.04	71.20±0.20
	50	97.09±0.34	96.35±0.87	91.20±0.55	38.40±8.79	83.28±24.75	82.50±0.95	70.45±0.44
HM-M	10	92.91±1.01	93.09±1.64	91.15±0.75	32.13±3.02	76.36±9.57	82.02±0.76	66.67±0.15
	20	93.92±0.53	93.56±1.04	89.02±1.49	34.54±2.56	75.40±9.92	82.23±0.73	65.49±0.24
	50	92.54±0.79	92.85±1.31	88.87±0.84	30.20±9.40	74.10±16.66	78.78±1.57	64.67±0.26

HM-A and HM-M are reported as mean ± standard deviation over five seeds

Table 19 Standard deviations of ablations

Study	Setting	Scope	HM-A	HM-M
MPP retrieval	Visual prompts only	Caltech101, Oxford Flowers, Food-101, FGVC-Aircraft	80.69±0.57	76.25±1.49
	Text prompts only	Caltech101, Oxford Flowers, Food-101, FGVC-Aircraft	81.19±0.67	76.69±1.47
	Visual + text prompts	Caltech101, Oxford Flowers, Food-101, FGVC-Aircraft	**81.24±0.71**	**76.95±0.76**
Small datasets	FGVC-Aircraft	imb = 10	**40.75±0.91**	32.15±2.40
		imb = 20	39.09±1.73	**34.58±2.78**
	EuroSAT	imb = 10	**83.54±7.70**	**74.56±7.57**
		imb = 20	80.53±6.11	71.15±5.89

Results are reported as mean ± standard deviation
Bold values indicate the best performance among the compared methods

Appendix D More Experiment Results

We provide more experiments under more imbalanced settings here, including base-to-new experiments under imbalance factor of 100/200, and 1/2/4/8-shot base-to-new experiments. Results for base-to-new with imbalance factors of 100/200 are jointly reported in Tables 15 and 16. Results for base-to-new with 1/2/4/8-shot are reported in Tables 21, 22, 23, 24.

For 100/200 imbalance factor base-to-new experiments, we can find that LAMP also yields the best accuracy in these settings. While most existing methods suffer from dramatic performance degradation under these severe settings, and some even perform worse than zero-shot CLIP under several datasets, LAMP shows its unique robustness, outperforming the second-best results by an average of 0.71% under imbalance ratio 100 and 0.52% under imbalance ratio 200. This highlights that the design of LAMP is strong enough to handle extreme circumstances.

For the 1/2/4/8-shot base-to-new experiments, LAMP consistently achieves the best average accuracy, with improvements of 1.71%, 1.84%, 1.25%, and 2.42% over the previous state-of-the-art methods. LAMP also performs well on integrated and challenging datasets such as ImageNet, DTD, and EuroSAT. On sparse datasets with limited cardinality, such as FGVC-Aircraft, the results show larger variance. As the number of shots increases from 1 to 8, all methods show clear accuracy gains, and the gain of LAMP is larger than those of the competing methods. Under the 1/2-shot settings, MMA performs worse than the text-only prompt methods CoOp and CoCoOp, which may be caused by the directly shared bottleneck between visual and textual tokens. This bottleneck is difficult to optimize when training data are highly limited.

Appendix E Usage of LLMs

LLMs are only for improving the writing of this paper.

Table 20 Standard deviations for the LBO calibration ablation

Setting	Metric	Avg.	Caltech101		OxfordPets		Food-101		FGVC-Aircraft		EuroSAT		ImageNet	
			10	20	10	20	10	20	10	20	10	20	10	20
with calibration	HM-A	**80.62±1.54**	**97.58±0.42**	**97.88±0.28**	**97.76±0.98**	**96.87±0.53**	**92.45±0.71**	**91.62±0.50**	**40.69±0.52**	**39.04±1.72**	**84.78±5.04**	**84.65±7.41**	**72.96±0.18**	**71.20±0.20**
	HM-M	**75.35±1.66**	**92.91±0.12**	**93.92±0.51**	**93.09±0.62**	**93.56±1.18**	**91.15±0.52**	**89.02±0.38**	**32.13±0.59**	**34.54±2.18**	**76.36±8.30**	**75.40±5.16**	**66.67±0.15**	**65.49±0.24**
without calibration	HM-A	79.22±2.98	96.86±0.42	97.55±0.28	97.30±0.98	96.54±0.53	91.67±0.71	90.55±0.50	38.45±0.52	37.04±1.72	82.07±5.04	80.05±7.41	72.18±0.20	70.40±0.20
	HM-M	73.49±3.17	92.50±0.12	92.73±0.51	92.69±0.62	92.83±1.18	90.22±0.52	87.83±0.33	29.75±0.59	31.98±2.18	70.73±9.23	69.49±5.16	65.99±0.34	65.10±0.37

Results are reported as mean ± standard deviation. The Avg. column is computed over all listed datasets and imbalance ratios
Bold values indicate the best performance among the compared methods

Table 21 Comparison between LAMP and other methods under 1-shot base-to-new settings

Method	Average	ImageNet	Caltech101	OxfordPets	StanfordCars	Flowers102
CoOp	67.56	66.33	92.60	90.37	67.43	77.53
CoCoOp	66.79	69.43	93.83	91.27	67.22	72.08
MaPLe	69.27	62.67	92.57	89.10	66.60	83.30
MMA	69.28	69.17	**92.90**	91.23	67.87	83.60
LAMP	**70.99**	**70.69**	91.81	**93.64**	**68.64**	**84.39**

Method	Food101	FGVCAircraft	SUN397	DTD	EuroSAT	UCF101
CoOp	84.33	21.37	66.77	50.23	54.93	71.23
CoCoOp	**85.65**	12.68	**68.33**	48.54	55.33	70.30
MaPLe	80.50	26.73	64.77	52.13	**71.80**	71.83
MMA	83.03	**28.73**	64.00	52.27	55.07	74.17
LAMP	81.64	27.02	64.19	**54.69**	68.24	**76.04**

We report the Harmonic Mean (HM) over base sets and new sets. The best and second-best results are marked with bold and underlined

Table 22 Comparison between LAMP and other methods under 2-shot base-to-new settings

Method	Average	ImageNet	Caltech101	OxfordPets	StanfordCars	Flowers102
CoOp	70.65	67.07	93.07	89.80	70.50	87.33
CoCoOp	67.65	69.78	94.82	92.64	68.37	75.79
MaPLe	72.58	65.10	93.97	90.87	71.60	88.93
MMA	72.08	**70.37**	94.00	91.97	71.77	90.30
LAMP	**74.42**	70.34	**95.53**	**94.64**	**72.46**	**91.24**

Method	Food101	FGVCAircraft	SUN397	DTD	EuroSAT	UCF101
CoOp	84.40	26.20	66.53	53.60	65.17	73.43
CoCoOp	**86.22**	15.06	**69.03**	52.17	46.74	73.51
MaPLe	81.47	30.90	67.10	55.50	**78.30**	74.60
MMA	82.50	**31.90**	67.17	56.90	59.80	76.17
LAMP	84.09	28.81	68.14	**62.99**	72.24	**78.16**

We report the Harmonic Mean (HM) over base sets and new sets. The best and second-best results are marked with bold and underlined

Table 23 Comparison between LAMP and other methods under 4-shot base-to-new settings

Method	Average	ImageNet	Caltech101	OxfordPets	StanfordCars	Flowers102
CoOp	74.02	68.73	94.40	92.57	74.47	92.17
CoCoOp	71.21	70.39	**94.98**	92.81	69.39	78.40
MaPLe	75.37	67.70	94.43	91.90	75.30	92.67
MMA	76.38	71.00	94.33	92.33	76.50	93.00
LAMP	**77.63**	**71.14**	93.94	**93.04**	**77.24**	**94.09**

Method	Food101	FGVCAircraft	SUN397	DTD	EuroSAT	UCF101
CoOp	84.47	30.83	69.97	58.70	70.80	77.10
CoCoOp	86.88	24.79	70.21	55.04	65.56	74.82
MaPLe	81.77	34.87	70.67	61.00	**84.50**	78.47
MMA	82.13	**37.57**	69.97	63.93	79.40	80.10
LAMP	**82.94**	36.36	**72.14**	**68.83**	83.54	**80.69**

We report the Harmonic Mean (HM) over base sets and new sets. The best and second-best results are marked with bold and underlined

Table 24 Comparison between LAMP and other methods under 8-shot base-to-new settings

Method	Average	ImageNet	Caltech101	OxfordPets	StanfordCars	Flowers102
CoOp	76.98	70.63	94.37	91.27	79.30	94.97
CoCoOp	72.96	70.63	95.04	93.45	70.44	84.30
MaPLe	78.89	70.30	95.20	92.57	79.47	<u>95.80</u>
MMA	<u>79.57</u>	<u>71.77</u>	<u>95.37</u>	<u>92.77</u>	<u>81.40</u>	**95.97**
LAMP	**81.99**	**72.83**	**97.26**	**93.67**	**93.77**	94.2

Method	Food101	FGVCAircraft	SUN397	DTD	EuroSAT	UCF101
CoOp	82.67	39.00	71.53	64.77	78.07	84.93
CoCoOp	<u>86.97</u>	26.61	70.84	58.89	68.21	73.32
MaPLe	83.60	<u>42.00</u>	**73.23**	66.50	<u>87.73</u>	92.33
MMA	83.00	**44.83**	<u>72.30</u>	<u>67.97</u>	86.47	<u>92.37</u>
LAMP	**87.82**	35.97	71.46	**71.69**	**90.08**	**93.14**

We report the Harmonic Mean (HM) over base sets and new sets. The best and second-best results are marked with bold and underlined

Appendix F Reproducibility

The code, training scripts, configuration files, and pretrained checkpoints are available at https://github.com/xuc865/lamp. See Table 25.

Table 25 Unified hyperparameter setting for LAMP

Config field	Subfield	base-to-new imb	base-to-new few-shot	cross-dataset / domain-generalization
dataloader	train:batch-size	32	32	32
	test:batch-size	100	100	100
	num_workers	8	8	8
	size	[224,224]	[224,224]	[224,224]
input	interpolation	bicubic	bicubic	bicubic
	pixel_mean	[0.4814,0.4578,0.4082]		
	pixel_std	[0.2686, 0.2613, 0.2757]		
	transforms	["random_resized_crop", "random_flip", "normalize"]		
	name	adamw	adamw	adamw
optim	lr	0.0003	0.0003	0.0003
	max_epoch	50	50	50
	lr_scheduler	cosine	cosine	cosine
	warmup_epoch	1	1	1
	warmup_type	constant	constant	constant
	warmup_cons_lr	1e-5	1e-5	1e-5
	print_freq	5	5	5
train	backbone.name	VIT-B/16	VIT-B/16	VIT-B/16
model	share_dim	32	32	32
trainer.prolt	share_layers	[6,7,8,9,10,11,12]		
	share_token_num	2	4	4
	extra_token_num	2	4	4
	prec	amp	amp	amp

The main B2N results use the same 50-epoch protocol

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Author Contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Xucong Wang. The first draft of the manuscript was written by Xucong Wang. Zhe Zhao, Liheng Yu, and Pengkun Wang revised the manuscript. Yang Wang provided overall guidance and supervision. All authors read and approved the final manuscript.

Data Availability The datasets analyzed during the current study are available in their respective public repositories: ImageNet https://image-net.org/index.php, Caltech101 http://www.vision.caltech.edu/Image_Datasets/Caltech101/101_ObjectCategories.tar.gz, SUN397 http://vision.princeton.edu/projects/2010/SUN/SUN397.tar.gz, DTD https://www.robots.ox.ac.uk/vgg/data/dtd/download/dtd-r1.0.1.tar.gz, EuroSAT http://madm.dfki.de/files/sentinel/EuroSAT.zip, UCF101 https://drive.google.com/file/d/10Jqome3vtUA2keJkNanAiFpgbyC9Hc2O/view?usp=sharing, OxfordPets https://www.robots.ox.ac.uk/\nobreakspace{}vgg/data/pets/data/images.tar.gz, StanfordCars http://ai.stanford.edu/\protect\unhbox\voidb@x\penalty\@M\jkrause/car196, Flowers102 https://www.robots.ox.ac.uk/vgg/data/flowers/102/102flowers.tgz, Food101 https://data.vision.ee.ethz.ch/cvl/datasets_extra/food-101/, FGVC-Aircraft https://www.robots.ox.ac.uk/\protect\unhbox\voidb@x\penalty\@M\vgg/data/fgvc-aircraft/archives/fgvc-aircraft-2013b.tar.gz, ImageNet-A https://github.com/hendrycks/natural-adv-examples, ImageNet-S https://github.com/HaohanWang/ImageNet-Sketch, ImageNet-R https://github.com/hendrycks/imagenet-r, and ImageNetV2 https://github.com/modestyachts/ImageNetV2.

Code Availability The source code and pre-trained models are available at https://github.com/xuc865/lamp.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethics Approval Not applicable. This study does not involve human participants or animals. The research utilizes publicly available benchmark datasets (ImageNet (Deng et al., 2009), Caltech101 (Fei-Fei et al., 2004), SUN397 (Xiao et al., 2010), DTD (Cimpoi et al., 2014), EuroSAT (Helber et al., 2019), UCF101 (Soomro et al., 2012), OxfordPets (Parkhi et al., 2012), StanfordCars (Krause et al., 2013), Flowers102 (Nilsback & Zisserman, 2008), Food101 (Bossard et al., 2014), FGVC-Aircraft (Maji et al., 2013), ImageNet-A (Hendrycks et al., 2021b), ImageNet-S (Wang et al., 2019), ImageNet-R (Hendrycks et al., 2021a), and ImageNetV2 (Recht et al., 2019)).

Consent to Participate Not applicable.

Consent for Publication Not applicable.

Code availability The source code and pre-trained models are available at https://github.com/xuc865/lamp.

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